

41. Page Lifecycle: DOMContentLoaded, load, beforeunload, unload

HTML page ke lifecycle me 3 important events hain:

Event	Kab	Kaam ka
DOMContentLoaded	HTML pura load, DOM tree ban gaya (images/styles abhi baaki ho sakte hain)	DOM nodes dhundhna, interface initialize karna
load	HTML + saare external resources (images, styles) load	Image sizes, styles applied hone ke baad ka kaam
beforeunload / unload	User page chhod raha hai	Unsaved changes ka confirm / stats bhejna

DOMContentLoaded

document par hota hai, sirf addEventListener se lagta hai (document.addEventListener nahi chalta).

```
js
document.addEventListener("DOMContentLoaded", ready);
```

Ye image ka wait **nahi** karta (isliye image ki size 0x0 dikh sakti hai).

Scripts ke saath

Browser jab `<script>` dekhta hai to pehle use **chalata** hai, tab aage DOM banata hai. To `DOMContentLoaded` un scripts ke **baad** hi aata hai.

Exceptions (jo `DOMContentLoaded` ko block nahi karte):

1. `async` attribute wale scripts
2. `document.createElement('script')` se dynamically bane scripts

Styles ke saath

External stylesheet DOM ko affect nahi karti, to `DOMContentLoaded` uska wait nahi karta. **Par** agar stylesheet ke baad script hai, to wo script stylesheet ke load hone tak ruka rehta hai (style-dependent cheezein padhne ke liye), aur `DOMContentLoaded` us script ka wait karta hai. To indirectly styles ka bhi wait hota hai.

Autofill

Firefox, Chrome, Opera (`DOMContentLoaded`) par form autofill karte hain. Agar bhaari scripts isse late karein, to login/password fields der se bharte hain.

`window.onload`

Poora page + saare resources load hone par. Yaha image sizes sahi milte hain.

```
js
window.onload = function() { ... }; // ya window.addEventListener('load', ...
```

`window.onunload`

User page chhodte waqt. Yaha aisa kaam karo jisme **der na lage** (jaise related popup band karna). Analytics bhejni ho to (`navigator.sendBeacon`):

```
js
window.addEventListener("unload", function() {
  navigator.sendBeacon("/analytics", JSON.stringify(analyticsData));
});
```

- POST request jaati hai, background me (page chhodne me deri nahi)
- Data limit 64kb
- Response nahi mil sakta
- `fetch` me `keepalive` flag bhi isi kaam ke liye hai.

Page chhodne ka transition **cancel** karna ho to `unload` me nahi, `beforeunload` me hota hai.

`window.onbeforeunload`

User page chhodne ya window band karne ki koshish kare to confirmation maangta hai.

```
js
window.onbeforeunload = function() { return false; };
```

- Purane browsers non-empty string return karne par wo message dikhte the, ab **custom message nahi** dikhta (misuse ki wajah se).
- `event.preventDefault()` yaha aksar **kaam nahi karta**. Iski jagah `event.returnValue = "..."` set karo:

```
js
```

```
window.addEventListener("beforeunload", (event) => {
  event.returnValue = "There are unsaved changes. Leave now?";
});
```

readyState

Agar `DOMContentLoaded` handler document ke load hone ke **baad** lagaya, to wo kabhi nahi chalega. Isliye `document.readyState` check karo:

- `"loading"`: document load ho raha hai
- `"interactive"`: document poora padha ja chuka (`DOMContentLoaded` se just pehle)
- `"complete"`: document + saare resources load (`window.onload` se just pehle)

```
js
if (document.readyState == 'loading') {
  document.addEventListener('DOMContentLoaded', work);
} else {
  work(); // DOM pehle se ready
}
```

`readystatechange` event bhi hai (state badalne par), par ab kam use hota hai.

Typical order

1. `readyState: loading`
2. `readyState: interactive`
3. `DOMContentLoaded`
4. `iframe onload`
5. `img onload`
6. `readyState: complete`
7. `window onload`

Summary

- `DOMContentLoaded`: DOM ready (scripts block karte hain, images ka wait nahi).
- `load`: sab kuch load (kam use, kyunki der lagti hai).
- `beforeunload`: leave confirm; `unload`: sirf simple kaam, analytics ke liye `sendBeacon`.
- `readyState` se pata karo ki document kis state me hai.